Health Center Management System

System Scenarios – QSHMS

Scenario 1: Patient Visit Management by Doctor

User: *Dr. Marku, a general physician.*

1. **Initiation**
   - Patient Arben enters Dr. Marku's consultation room after triage.
   - Dr. Marku opens the consultation module in QSHMS.
2. **Interaction with System**
   - He reviews Arben's medical history, allergies, and triage vitals.
   - During consultation, he records symptoms, makes a diagnosis (hypertension), and prescribes medication.
   - The system automatically checks for drug interactions based on Arben's history.
3. **Hospital Staff Interaction**
   - Dr. Marku orders blood tests for Arben; the system notifies the lab technician.
   - He schedules a follow-up in 2 weeks; QSHMS sends an SMS reminder to Arben.
   - If Arben's blood pressure is critically high, the system flags the case as "Urgent – Nurse Follow-up Required".

Scenario 2: Patient Registration & Triage

User: *Fatmira, a receptionist.*

1. **Initiation**
   - A new patient, Genti, walks in without an appointment, complaining of high fever.
2. **Interaction with System**
   - Fatmira opens the Patient Registration module in QSHMS.
   - She enters Genti's personal details, symptoms, and emergency contact.
   - The system auto-generates a unique QSHMS Patient ID and checks for duplicates.
3. **Hospital Staff Interaction**
   - Fatmira selects "Triage Required"; the system instantly notifies the triage nurse.
   - The nurse measures vital signs (temperature, BP, pulse) and records them directly into QSHMS.
   - The system places Genti in the priority queue based on his fever (38.9°C).

Scenario 3: Lab Test Ordering & Results Viewing

User: *Ardian, a lab technician.*

1. **Initiation**
   - Dr. Marku orders a complete blood count (CBC) and fasting glucose for patient Lulja via QSHMS.
2. **Interaction with System**
   - Ardian sees the pending request in his "Laboratory Requests" dashboard.
   - He scans Lulja's QSHMS barcode from her digital profile.
   - He performs the tests and enters the results into the system.
3. **Hospital Staff Interaction**
   - QSHMS flags high glucose (180 mg/dL) as abnormal and sends an automatic alert to Dr. Marku.
   - Dr. Marku reviews the results remotely and prescribes diet changes and medication.
   - Ardian marks the request as "Completed – Reviewed by Clinician".

Scenario 4: Appointment Booking & Insurance Verification

User: *Blerina, a patient with private health insurance.*

1. **Initiation**
   - Blerina logs into the QSHMS patient portal via her smartphone.
2. **Interaction with System**
   - She selects "Book Appointment" → Dr. Marku → Wednesday at 9:00 AM.
   - The system shows available slots; she chooses one and enters her insurance policy number.
   - QSHMS automatically verifies coverage via API with the insurance company's system.
3. **Hospital Staff Interaction**
   - Insurance approval is returned within 10 seconds (20% copay).
   - Blerina confirms the booking; QSHMS sends SMS and email confirmation with a calendar link.
   - The insurance company receives a real-time pre-authorization request for the visit.

Scenario 5: Executive Dashboard & Compliance Monitoring

User: *Mr. Leka, Director of the Health Center.*

1. **Initiation**
   - Mr. Leka logs into the QSHMS Executive Dashboard every Monday morning.
2. **Interaction with System**
   - He views live KPIs: Average patient wait time 18 minutes (target <30 min), patient no-show rate 8%, doctor performance metrics (Dr. Marku's satisfaction score: 95%).
   - He drills down into department-wise performance and staff attendance logs.
3. **Hospital Staff Interaction**
   - Mr. Leka sees Dr. Marku's high satisfaction and sends him a recognition note via QSHMS.

- The system flags 3 incomplete security audit logs; he assigns the IT administrator to review them.
- He generates a weekly compliance report directly from QSHMS for the Ministry of Health.

Scenario 6: Search for Existing Patient

User: *Fatmira, receptionist.*

1. **Initiation**
   - A returning patient, Mirela, forgot her QSHMS ID card.
2. **Interaction with System**
   - Fatmira opens the Search Patient module.
   - She types Mirela's last name and birth date.
   - The system returns one matching record within 2 seconds.
3. **Hospital Staff Interaction**
   - Fatmira verifies the phone number with Mirela and updates her contact info.
   - She books a same-day appointment with Dr. Marku.
   - The system logs the search action for audit purposes.

Scenario 7: Cancel Appointment

User: *Fatmira, receptionist.*

1. **Initiation**
   - Patient Kreshnik calls to cancel his 2:00 PM appointment due to emergency.
2. **Interaction with System**
   - Fatmira searches for Kreshnik's appointment by name and date.
   - She selects "Cancel Appointment" and chooses reason: "Patient requested cancellation".
3. **Hospital Staff Interaction**
   - System changes status to "Cancelled" and frees the time slot.
   - An automatic notification is sent to Dr. Marku's calendar.
   - QSHMS asks Fatmira if she wants to send a cancellation confirmation SMS to Kreshnik – she clicks Yes.

Scenario 8: Update Patient Information

User: *Fatmira, receptionist.*

1. **Initiation**
   - Patient Shpresa moved to a new address and changed her phone number.
2. **Interaction with System**
   - Fatmira searches for Shpresa's record.
   - She edits the address, phone, and adds a new emergency contact.

- The system validates the phone number format and saves a timestamped change log.

3. **Hospital Staff Interaction**
   - Fatmira prints an updated patient information sheet for Shpresa to sign.
   - The billing department is automatically notified of the address change for future invoices.
   - Shpresa receives an SMS confirming her record was updated (optional privacy setting).

Scenario 9: Add Diagnosis After Consultation

User: *Dr. Marku.*

1. **Initiation**
   - Patient Lulja returns for a follow-up on her lab results.
2. **Interaction with System**
   - Dr. Marku opens Lulja's medical record and clicks "Add Diagnosis".
   - He enters "Type 2 Diabetes – newly diagnosed" and adds clinical notes.
   - The system suggests linking the diagnosis to the recent lab order (glucose 180 mg/dL).
3. **Hospital Staff Interaction**
   - Dr. Marku saves the diagnosis; it appears in Lulja's medical history.
   - The system automatically creates a task for a dietitian to consult Lulja.
   - A summary is added to the patient's next appointment preparation notes.

Scenario 10: Create Prescription with Allergy Check

User: *Dr. Marku.*

1. **Initiation**
   - After diagnosing Lulja with diabetes, Dr. Marku decides to prescribe Metformin.
2. **Interaction with System**
   - He selects "Create Prescription" and searches for Metformin.
   - The system checks Lulja's allergy list – no known allergies to Metformin.
   - He enters dosage (500 mg twice daily) and duration (30 days).
3. **Hospital Staff Interaction**
   - The prescription is saved and digitally signed by Dr. Marku.
   - QSHMS sends the prescription to the on-site pharmacy for preparation.
   - Lulja receives an SMS: "Your prescription is ready for pickup."

Scenario 11: View Doctor Schedule

User: *Fatmira, receptionist.*

1. **Initiation**
   - A patient wants to book an appointment with Dr. Marku next week.

2. **Interaction with System**
   - Fatmira opens "View Doctor Schedule" and selects Dr. Marku.
   - She chooses date range: Monday to Friday next week.
   - The system displays booked slots (red) and free slots (green).
3. **Hospital Staff Interaction**
   - Fatmira sees that Thursday afternoon has three free slots.
   - She books the patient for Thursday at 3:00 PM.
   - The schedule updates in real time, blocking that slot for other receptionists.

Scenario 12: Process Payment

User: *Arta, cashier.*

1. **Initiation**
   - Patient Genti finishes his consultation and lab tests; he approaches the payment counter.
2. **Interaction with System**
   - Arta retrieves Genti's invoice by scanning his QSHMS ID barcode.
   - The system shows total amount: 2,500 ALL (consultation + CBC).
   - Genti pays by card; Arta selects "Card Payment" and enters the amount.
3. **Hospital Staff Interaction**
   - The system connects to the payment terminal; transaction approved.
   - Invoice status changes to "Paid", and a receipt is printed.
   - The accounting department sees the payment in real-time dashboard.

Scenario 13: Generate Invoice Automatically

User: *System (automated) + Arta (cashier, if needed).*

1. **Initiation**
   - Dr. Marku marks patient Mirela's consultation as "Completed".
2. **Interaction with System**
   - QSHMS collects all billable items: consultation fee, ECG, and a wound dressing.
   - It calculates total (3,200 ALL) and generates an invoice with unique number INV-2025-0421.
3. **Hospital Staff Interaction**
   - The invoice status is "Unpaid" and appears in Mirela's patient portal.
   - Arta can view it under "Pending Payments".
   - Mirela receives an email: "Your invoice is ready. Pay online or at the counter."

Scenario 14: Request Lab Test

User: *Dr. Marku.*

1. **Initiation**
   - Patient Arben's symptoms suggest possible thyroid disorder.
2. **Interaction with System**
   - Dr. Marku opens "Request Lab Test" from Arben's record.
   - He selects TSH, T3, T4, and adds note: "Check for hypothyroidism."
   - The system verifies that these tests are available at the center.
3. **Hospital Staff Interaction**
   - The lab order appears instantly in Ardian's (lab technician) dashboard.
   - A barcode is generated for sample tubes.
   - Arben receives an SMS: "Lab tests ordered. Please go to the lab room."

Scenario 15: Enter Lab Results

User: *Ardian, lab technician.*

1. **Initiation**
   - Ardian completes the CBC for patient Lulja.
2. **Interaction with System**
   - He searches for pending lab orders and selects Lulja's CBC request.
   - He enters numerical results: WBC, RBC, hemoglobin, platelets.
   - The system validates that all required fields are filled.
3. **Hospital Staff Interaction**
   - Results are saved and marked "Completed".
   - QSHMS automatically notifies Dr. Marku via in-app alert and email.
   - If any value is outside normal range, the system adds a yellow flag.

Scenario 16: View Patient Medical History

User: *Dr. Marku.*

1. **Initiation**
   - Patient Shpresa returns after 6 months with a new complaint of joint pain.
2. **Interaction with System**
   - Dr. Marku opens Shpresa's record and clicks "Medical History".
   - The system retrieves past diagnoses (osteoarthritis, GERD), prescriptions, and lab results in reverse chronological order.
3. **Hospital Staff Interaction**
   - Dr. Marku notes that Shpresa had a similar episode 1 year ago.
   - He adds the new complaint and links it to the previous diagnosis.
   - The system suggests checking for medication interactions with her existing NSAID prescription.

Scenario 17: Refer to Specialist

User: *Dr. Marku.*

1. **Initiation**
   - Patient Arben's hypertension is uncontrolled despite two medications; cardiology referral needed.
2. **Interaction with System**
   - Dr. Marku selects "New Referral" from Arben's record.
   - He chooses "Cardiology" from the specialist list, enters reason, and attaches recent BP logs.
   - The system generates a formatted referral letter.
3. **Hospital Staff Interaction**
   - Dr. Marku prints the letter for Arben to take to the specialist.
   - QSHMS records the referral in Arben's history with date and reason.
   - An optional task is created for the receptionist to follow up on referral outcome.

Scenario 18: Manage Inventory

User: *Endrit, system administrator.*

1. **Initiation**
   - Monthly stock check shows low levels of amoxicillin and gloves.
2. **Interaction with System**
   - Endrit opens "Inventory Management".
   - He searches for "amoxicillin 500mg" and sees current stock = 120 tablets (threshold = 200).
   - He adds a new shipment of 500 tablets and updates quantity to 620.
3. **Hospital Staff Interaction**
   - The system automatically clears the reorder alert for amoxicillin.
   - The pharmacy staff sees the updated stock in real time.
   - Endrit exports a low-stock report for the procurement team.

Scenario 19: Record Vital Signs

User: *Valbona, triage nurse.*

1. **Initiation**
   - Patient Genti is called from the waiting area for triage.
2. **Interaction with System**
   - Valbona selects Genti from "Today's Appointments" list.
   - She enters BP (130/85), temperature (38.9°C), pulse (102 bpm), respiration (20).
   - The system validates ranges and flags temperature as high (warning icon).

3. **Hospital Staff Interaction**
   - The vital signs are saved with a timestamp and become visible to Dr. Marku immediately.
   - QSHMS updates the waiting room display: Genti moved to "With Nurse".
   - If any vital is critical (e.g., BP >180/120), the system sends an urgent alert to the doctor.

Scenario 20: Send Appointment Reminder

User: *System (scheduled job).*

1. **Initiation**
   - Time-based trigger runs every evening at 8:00 PM.
2. **Interaction with System**
   - QSHMS queries appointments for the next day (8:00 AM – 8:00 PM).
   - For each appointment, it retrieves patient contact info (SMS number, email).
   - It composes a message: "Reminder: Dr. Marku tomorrow at 10:00 AM. Reply 1 to confirm."
3. **Hospital Staff Interaction**
   - SMS and email are sent to 42 patients.
   - The system logs each reminder (timestamp, delivery status).
   - If a patient has no valid contact, the system flags the appointment for receptionist follow-up.